

DATABASE_ER

Helix OTA — Database Entity-Relationship Diagram

Overview

This ER diagram shows all persistent entities in the Helix OTA PostgreSQL database, their columns, data types, and relationships. The schema supports multi-tenant operations, full audit logging, and campaign/rollout management.

Diagram

```
erDiagram
    ORGANIZATIONS {
        uuid id PK
        string name
        string slug UK
        jsonb settings
        timestamp created_at
        timestamp updated_at
    }

    USERS {
        uuid id PK
        uuid organization_id FK
        string email UK
        string password_hash
        string role "admin|operator|viewer"
        string display_name
        boolean active
        timestamp last_login
        timestamp created_at
        timestamp updated_at
    }

    API_KEYS {
        uuid id PK
        uuid organization_id FK
```

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        uuid user_id FK
        string key_hash UK
        string name
        string[] scopes "read,write,deploy"
        timestamp expires_at
        timestamp last_used
        timestamp created_at
    }

    DEVICE_GROUPS {
        uuid id PK
        uuid organization_id FK
        string name
        string description
        jsonb filter_rules "hw_revision, region, etc."
        timestamp created_at
        timestamp updated_at
    }

    DEVICES {
        uuid id PK
        uuid organization_id FK
        uuid device_group_id FK
        string device_id UK "unique hardware ID"
        string hw_revision
        string current_build
        string oem_version
        string android_version
        string serial_number
        string region
        jsonb metadata
        string status "online|offline|unknown"
        timestamp last_seen
        timestamp created_at
        timestamp updated_at
    }

    ARTIFACTS {
        uuid id PK
        uuid organization_id FK
        string name
        string version
        string target_hardware
        string source_build_min
        string source_build_max
        string artifact_type "full|delta"
        string hash_sha256
        string signature_ed25519
        bigint size_bytes
        string minio_bucket
        string minio_key
        string status "uploading|validating|ready|rejected|
quarantined"
        string rejection_reason

```

```

    jsonb metadata
    uuid delta_source_artifact_id FK
    timestamp created_at
    timestamp updated_at
}

CAMPAIGNS {
    uuid id PK
    uuid organization_id FK
    uuid artifact_id FK
    string name
    string description
    string strategy "canary|linear|forced"
    jsonb rollout_stages "[5,10,30,50,100]"
    jsonb success_thresholds "[95,95,97,98]"
    jsonb min_duration_minutes "[30,30,60,120]"
    jsonb device_filter "targeting rules"
    uuid device_group_id FK
    string urgency "low|normal|high|critical"
    boolean allow_deferral
    integer max_deferrals
    string status "draft|active|paused|halted|completed|
rolled_back"
    timestamp started_at
    timestamp completed_at
    uuid created_by FK
    timestamp created_at
    timestamp updated_at
}

ROLLOUTS {
    uuid id PK
    uuid campaign_id FK
    integer stage_index
    integer percentage "5|10|30|50|100"
    bigint total_devices
    bigint succeeded_count
    bigint failed_count
    bigint in_progress_count
    bigint pending_count
    string status "pending|active|completed|rolled_back"
    timestamp started_at
    timestamp completed_at
    timestamp created_at
}

DEVICE_ROLLOUTS {
    uuid id PK
    uuid rollout_id FK
    uuid device_id FK
    string status "selected|notified|downloading|installing|
succeeded|failed|rolled_back|deferred"
    integer deferral_count
    string last_error

```

```

        timestamp selected_at
        timestamp notified_at
        timestamp download_started_at
        timestamp download_completed_at
        timestamp install_started_at
        timestamp install_completed_at
        timestamp reported_at
        timestamp created_at
        timestamp updated_at
    }

    DEVICE_REPORTS {
        uuid id PK
        uuid device_id FK
        uuid device_rollout_id FK
        uuid campaign_id FK
        string state "checking|downloading|verifying|installing|
rebooting|committing|succeeded|failed"
        integer progress_percent
        string error_code
        string error_message
        jsonb details "diagnostics"
        string reported_build
        timestamp reported_at
    }

    AUDIT_LOG {
        uuid id PK
        uuid organization_id FK
        uuid actor_id FK "nullable (system actions)"
        string actor_type "user|api_key|system"
        string action "campaign.create|artifact.upload|
rollout.pause..."
        string resource_type "campaign|artifact|device|rollout"
        uuid resource_id
        jsonb old_value
        jsonb new_value
        string ip_address
        string user_agent
        timestamp created_at
    }

    NOTIFICATION_RULES {
        uuid id PK
        uuid organization_id FK
        string name
        string event_type "rollout.paused|device.failed|
artifact.rejected"
        jsonb conditions "threshold, device_count..."
        string[] channels "email|webhook|dashboard"
        jsonb channel_config "emails, webhook_url..."
        boolean active
        timestamp created_at
        timestamp updated_at
    }

```

```

}

SIGNING_KEYS {
    uuid id PK
    uuid organization_id FK
    string key_id UK
    string public_key
    string algorithm "ed25519"
    boolean active
    timestamp expires_at
    timestamp revoked_at
    timestamp created_at
}

||--o{ ORGANIZATIONS : USERS
||--o{ ORGANIZATIONS : DEVICE_GROUPS
||--o{ ORGANIZATIONS : API_KEYS
||--o{ ORGANIZATIONS : SIGNING_KEYS
||--o{ ORGANIZATIONS : NOTIFICATION_RULES
||--o{ ORGANIZATIONS : AUDIT_LOG
||--o{ ORGANIZATIONS : DEVICES
||--o{ ORGANIZATIONS : ARTIFACTS
||--o{ ORGANIZATIONS : CAMPAIGNS
USERS ||--o{ API_KEYS : "authenticates with"
USERS ||--o{ CAMPAIGNS : "creates"
USERS ||--o{ AUDIT_LOG : "performs actions"
DEVICE_GROUPS ||--o{ DEVICES : "contains"
DEVICE_GROUPS ||--o{ CAMPAIGNS : "targets"
DEVICES ||--o{ DEVICE_ROLLOUTS : "receives"
DEVICES ||--o{ DEVICE_REPORTS : "submits"
ARTIFACTS ||--o{ CAMPAIGNS : "deployed in"
ARTIFACTS |o--o{ ARTIFACTS : "delta_source"
CAMPAIGNS ||--o{ ROLLOUTS : "has stages"
ROLLOUTS ||--o{ DEVICE_ROLLOUTS : "assigns to devices"
DEVICE_ROLLOUTS ||--o{ DEVICE_REPORTS : "generates"
CAMPAIGNS ||--o{ DEVICE_REPORTS : "tracked by"
SIGNING_KEYS ||--o{ ARTIFACTS : "signs"

```

Key Relationships

Relationship	Type	Description
Organization → Users	One-to-Many	Multi-tenant isolation; all entities scoped to org
Organization → Devices	One-to-Many	Devices belong to an organization
Device Group → Devices	One-to-Many	Logical grouping for targeting
Artifact → Campaigns	One-to-Many	An artifact can be deployed in multiple campaigns

Relationship	Type	Description
Artifact → Artifact (delta)	Self-referencing	Delta patches reference their source artifact
Campaign → Rollouts	One-to-Many	A campaign has multiple rollout stages
Rollout → Device Rollouts	One-to-Many	Each rollout stage assigns devices
Device Rollout → Device Reports	One-to-Many	Device submits multiple progress reports
Signing Key → Artifacts	One-to-Many	Keys used to verify artifact signatures
User → Audit Log	One-to-Many	All user actions are audited

Indexes (Critical)

Table	Index	Purpose
devices	idx_devices_org_hw_build	Fast campaign targeting queries
devices	idx_devices_device_id	Lookup by hardware ID
device_reports	idx_reports_device_campaign	Fetch reports for a device+campaign
device_reports	idx_reports_state	Aggregate state counts for dashboards
device_rollouts	idx_rollouts_status	Find devices in specific rollout states
audit_log	idx_audit_org_created	Time-range audit queries per org
campaigns	idx_campaigns_status	Find active campaigns for scheduler